

Bunching at a Moving Threshold: Firm Responses to Romania’s Micro-Enterprise Tax, 2014–2025

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Abstract

Romanian companies below a euro-denominated revenue ceiling pay 1 to 3 percent of revenue instead of 16 percent of profit. The ceiling moved six times between 2013 and 2026. Using all annual financial statements, 9.6 million firm-years, we show three things. First, reported revenue spikes just below the ceiling in every year, the spike follows the ceiling as it moves, and within a fixed euro ceiling it sits within a few thousand lei of the ceiling’s value at the previous year-end exchange rate and moves with the rate. Second, because the ceiling moved, the density at each notch in years when no ceiling was there is a counterfactual that needs no polynomial. At 77 placebo locations with no notch this difference-in-bunching estimator gives excess mass centred on zero with a standard deviation of 0.15; at every actual notch it gives excess mass of 2 to 8 percent of firms within 20 percent of the ceiling, yet only 0.06 to 0.30 percent of all filers: bunching is large locally and trivial in aggregate, and the regime’s fiscal cost lies with the intramarginal firms. Third, the response is concentrated in high-margin service sectors, reported margins move in the predicted direction when firms are switched between regimes, a 2023 employee condition doubled the rate at which zero-employee firms took on one employee in the year it was enacted, and bunching is correlated with, though not shown to be caused by, the registration of sibling firms.

*Draft of September 20, 2026. Data: Ministry of Finance, ANAF and ONRC open data via data.gov.ro. Replication package (pipeline, harmonised panel with hashed identifiers, legal timeline): <https://github.com/simdenis/romania-micro-enterprise-bunching>, data files on Zenodo, doi:10.5281/zenodo.22863290; it contains no names, birth dates or addresses. Drafting and code were produced with the assistance of a large language model (Anthropic Claude), including the initial text of every section and the analysis scripts; all analysis choices, the verification of legal references against primary sources and of numerical claims against the data, and the final text are the author’s responsibility. Comments welcome.

1 Introduction

Dual-regime business tax systems, which tax small firms on turnover and larger firms on profit, exist in most emerging economies and several EU member states (Sharma et al., 2025; Kanbur and Keen, 2014; Dharmapala et al., 2011). The size threshold between the regimes is a notch: crossing it changes the tax base from revenue to profit and, for a firm whose margin exceeds the ratio of the two rates, raises the tax bill discretely. Responses to such notches have been estimated at VAT thresholds in the United Kingdom (Liu et al., 2021), Finland (Harju et al., 2019) and Spain (Almunia and López-Rodríguez, 2018), at turnover-tax and corporate-tax discontinuities in Pakistan (Kleven and Waseem, 2013), Armenia (Asatryan and Peichl, 2016), South Africa (Boonzaaier et al., 2019) and Costa Rica (Bachas and Soto, 2021). In most of these settings the threshold moves rarely or only with indexation, so the counterfactual density at the notch is extrapolated from the density away from it, usually with a polynomial. Blomquist et al. (2021) show that this extrapolation is not identified without shape restrictions and that the estimates depend on them. Best and Kleven (2018) use the introduction of a notch to compare the same location before and after, and Marx (2024) exploits panel variation.

Romania is a setting in which the threshold moves. Its micro-enterprise regime taxes revenue at 1 to 3 percent for companies with annual revenue below a euro-denominated ceiling, and the ceiling changed six times between 2013 and 2026: €65,000 until 2015, €100,000 in 2016, €500,000 in 2017, €1,000,000 from 2018 to 2022, €500,000 in 2023 and 2024, €250,000 in 2025 and €100,000 from 2026. Every location that was ever a notch was, in other years, an ordinary point of the revenue distribution. We use that variation descriptively and econometrically, and we use the accompanying changes in side conditions, the employee requirement and the linked-enterprise rule, to look at the channels.

The paper makes three points. First, the spike in reported revenue follows the legal ceiling up and down and, within a fixed euro ceiling, tracks the lei value at the previous year-end exchange rate to within a few thousand lei a year. That settles the round-number objection: a spike at 4,779,300 lei that moves to 4,869,400 lei the next year is not an accounting convention. When a future ceiling has been enacted, or announced and then withdrawn, a second spike appears at it. Second, the density at each notch location in years without a threshold there provides a counterfactual under an assumption we state and test. At 77 placebo locations with no notch the estimator gives excess mass with mean 0.01, standard deviation 0.15 and a maximum absolute value of 0.55; at each actual notch it gives excess mass of 2 to 8 percent of the firms within 20 percent of the ceiling, or 0.7 to 3.0 in the excess-mass statistic, with block-bootstrap standard errors of 0.07 to 0.16. In aggregate, however, the excess is 0.06 to 0.30 percent of all filers. Bunching is large relative to the local density and small relative to the economy; the fiscal cost of the

regime, and the policy debate around it (Ban and Buciu, 2023; IMF, 2025; Consiliul Fiscal, 2024), concern the intramarginal firms, not the threshold. Third, the channels. Excess mass is two to three times higher in professional, administrative, health and information services than in trade and transport, and bunchers in those sectors report margins of 30 to 40 percent. When the 2018 ceiling increase moved firms onto the revenue tax their reported margins rose by 0.7 points relative to firms left on profit tax, and when the 2023 cut moved firms back margins fell by 1 to 2 points; the regime changes what firms report, not only where they sit, as Bachas and Soto (2021) find for Costa Rica. A 2023 employee condition doubled the rate at which zero-employee firms took on exactly one employee in the year it was enacted. Firms at the ceiling are more likely to have a newly registered sibling firm, but combined revenue of commonly administered firms does not bunch, so we treat splitting as a correlation. Section 6 takes each channel in turn.

Section 2 describes the regime, Section 3 the data, Section 4 the estimator, Section 5 the main results and Section 6 the channels. Section 7 concludes.

2 Institutional background

The micro-enterprise income tax (*impozitul pe veniturile microîntreprinderilor*), Title III of the Fiscal Code (Law 227/2015) and before 2016 Title IV¹ of Law 571/2003, replaces the 16 percent profit tax with a tax on revenue for companies meeting a set of conditions, of which the main one is that revenue in the previous fiscal year was below a ceiling set in euro. Table 1 gives the ceiling and rates and Table 2 the side conditions by year of revenue; Appendix A gives the acts.

Three features matter for what follows. The first is timing. Under the in-year rule, a micro-enterprise whose cumulated revenue exceeds the ceiling during year t owes profit tax from that quarter (for 2013 to 2015, for the whole year), the ceiling being converted at the BNR rate at the close of year $t - 1$. Under the year-end rule, status for year $t + 1$ is set by revenue at 31 December of year t , converted at the rate at the close of year t . When the ceiling is unchanged the two rules place the notch a few thousand lei

Table 1: Micro-enterprise regime: ceiling and rates by year of revenue

Revenue year	Ceiling (EUR)	Rate on revenue
2013–2015	65,000	3%
2016	100,000	1% (≥ 2 employees), 2% (1), 3% (none)
2017	500,000 (from 1 Feb.)	1% (≥ 1 employee), 3% (none)
2018–2022	1,000,000	1% (≥ 1 employee), 3% (none)
2023	500,000	1%
2024	500,000	1% up to €60,000; 3% above or in listed activities
2025	250,000	1% up to €60,000; 3% above or in listed activities
2026	100,000	1%

Table 2: Micro-enterprise regime: side conditions by year of revenue

Years	Conditions
2013–2015	Mandatory (opt-out for new firms with capital $\geq$ €25,000). Consulting and management revenue at most 20% (from 2014). Exceeding the ceiling in-year re-taxed the whole year.
2016–2017	Mandatory, opt-out for capital $\geq$ 45,000 lei (from 2017). In-year exit from the quarter of crossing. Consulting cap retained.
2018–2022	Mandatory, opt-out for capital $\geq$ 45,000 lei with two employees (from April 2018). Consulting cap abolished.
2023	Optional; exit permanent. At least one employee. Consulting cap 20%. A person holding over 25% may be a shareholder in at most three micro-enterprises. Hospitality firms exempt from the ceiling.
2024	At most one micro-enterprise per person holding over 25%. Ceiling tested on linked enterprises combined. Consulting cap reformulated.
2025	Consulting cap abolished. Status for 2026 tested on 2025 revenue against €100,000.
2026	Base changed to accounting turnover including linked enterprises.

apart, and the in-year rule, which uses the earlier and lower rate, is the binding one in a depreciating currency. When a cut has been enacted for the following year a firm faces two notches. This happened in 2022, when Ordinance 16/2022 of July 2022 set €500,000 for 2023 status based on 2022 revenue, before Law 370/2022 of 20 December 2022 deferred the revenue test to 2023 revenue while keeping the employee and ownership conditions at 31 December 2022; and in 2025, when Ordinance 156/2024 of 31 December 2024 set €250,000 for 2025 and €100,000 for 2026 status based on 2025 revenue.

The second feature is the tax base. The ceiling is tested on the taxable base of Article 53, which excludes a short list of items such as subsidies, provision reversals and, from 2021, dividends received. The statements report total revenue, which is weakly larger; for most small firms the excluded items are zero, but where they are not the firm’s position relative to the ceiling is overstated, which biases the bunching region slightly to the right in our data. From 2026 the base is accounting turnover.

The third is the set of side conditions in Table 2. The employee condition is met by a full-time contract, part-time contracts summing to a full norm, or a paid mandate contract; linked enterprises are defined by holdings over 25 percent; and the listed activities taxed at 3 percent regardless of revenue in 2024 and 2025 are software, hospitality, legal and medical services. The consulting and management cap of 20 percent, in force in 2014–2017 and 2023, is a margin the statements do not let us observe.

The VAT small-enterprise exemption is a separate notch on turnover as defined in Article 310(2), set in lei: 220,000 lei until March 2018, 300,000 lei from 1 April 2018 (Law 72/2018) and 395,000 lei from 1 September 2025 (Ordinance 22/2025). In 2024 the lei value of the €60,000 rate boundary lies within 1,500 lei of 300,000, so the two cannot be separated; in 2025 the VAT threshold moved away and the boundary can be observed on

Table 3: Statement panel, 2014–2025

Year	Firm-years	Revenue > 0	Median revenue (k lei)	EUR/RON, 31 Dec
2014	676,868	495,914	124	4.4821
2015	681,821	505,212	141	4.5245
2016	700,249	526,282	146	4.5411
2017	732,051	553,283	151	4.6597
2018	757,929	579,309	162	4.6639
2019	792,376	616,900	173	4.7793
2020	815,779	636,361	162	4.8694
2021	781,470	624,795	188	4.9481
2022	862,564	693,309	207	4.9474
2023	890,836	716,718	223	4.9746
2024	968,594	769,620	226	4.9741
2025	918,232	738,762	239	5.0985

Source: Ministry of Finance annual financial statements (micro-entity and full forms) via data.gov.ro; BNR year-end reference rate. Median revenue is over firms with positive revenue. 32 duplicated firm-years (30 in 2016) removed; about 600 firm-years per year with negative revenue are excluded from all windows.

its own (Section 6.1).

3 Data

Financial statements. The Ministry of Finance publishes, for each fiscal year, the main indicators of the annual financial statements of every company, identified by fiscal code (CUI) and CAEN activity code: balance-sheet totals, net turnover, total revenue, total expenses, gross and net profit or loss, and average headcount. Two forms exist, the micro-entity form and the full form. Their layout is stable from 2016; the 2014 and 2015 files use an older column set, which we harmonise. The files for 2021, 2023 and 2025 are early uploads made a few weeks after the filing deadline, while the others are later re-issues, so coverage of late filers differs by year; about 40,000 small firms appear in 2020 and 2022 but not 2021. Table 3 summarises the panel: 9.58 million firm-years after removing 32 duplicated records, 1.56 million firms. Total revenue is the variable used for the micro-enterprise notch; net turnover is used for the VAT threshold.

Tax regime. The statements do not record the regime. We construct it in two ways. First, gross minus net profit is the income tax accrued; dividing it by revenue gives roughly 1, 2 or 3 percent for micro-enterprises, and dividing it by pre-tax profit gives roughly 16 percent for profit-tax firms. The difference comprises the profit tax, the micro-enterprise tax, the specific tax on hospitality firms in 2017–2022 and other taxes on the result, net of sponsorship and cash-register credits, so ratios such as 0.8 or 2.4 percent occur

Table 4: Tax regime inferred from the 2023 statements versus the ANAF tax-vector flag (June 2023)

Regime inferred from statements	ANAF: micro	ANAF: profit	ANAF: other/none	Not in ANAF
Micro (tax $\approx$ 1–3% of revenue)	277,265	19,677	312	7,102
Micro (tax $>$ 0, pre-tax loss)	103,544	14,248	87	1,966
Profit tax ($\approx$ 16% of profit)	4,421	92,503	64	2,045
No tax paid	34,486	220,021	273	10,942
Ambiguous	16,047	30,596	60	437
Other	19,272	34,966	158	344

ANAF flags: IMP120 = registered for micro-enterprise income tax, IMP100 = registered for profit tax. Where both measures are decisive, they agree for 92.5% of firms.

and a firm with positive tax and a pre-tax loss is most likely, but not certainly, a micro-enterprise. Second, the ANAF taxpayer registry, a snapshot dated 31 August 2023, carries one column per tax a firm is registered for; in the dataset’s column dictionary the column for obligation 120 is the micro-enterprise income tax and the column for obligation 100 the profit tax. Table 4 cross-tabulates the two for 2023. Of firms flagged for the micro-enterprise tax, 84 percent show a micro-enterprise tax pattern and 1 percent a profit-tax pattern; of firms flagged for profit tax, 22 percent show the profit-tax pattern, 8 percent a micro pattern, and half paid no tax, which is what loss-making and dormant profit-tax firms look like. Where both measures are decisive, 57 percent of firms, they agree in 92.5 percent of cases; 2.6 percent of 2023 filers are not in the registry. The inference identifies micro-enterprises well and profit-tax firms poorly, so we use the registry flag where regime membership matters and the statement-based measure only as a check.

Employees. Average headcount is blank for a quarter of small firms and reported as zero for a further 5 percent. We treat blanks as zero; among micro-enterprises in 2017 to 2022, when the 3 percent rate applied to firms without employees, three quarters of blank-headcount firms paid the 3 percent rate. The field is integer-valued in every record, and it behaves as an indicator of having had an employee at some point in the year rather than as a time-weighted average: of firms that go from zero to one employee in 2022, 12 percent paid the no-employee rate for most of that year, which a proper average would have rounded to zero. Late-year hires are therefore counted, and the 2022 timing below is not an artefact. The legal condition is met by a full-time contract, by part-time contracts summing to a full norm, or by a paid mandate contract, none of which the field distinguishes, so headcount is a proxy for the legal test in both directions.

Trade register. The National Trade Register Office publishes snapshots of all registered firms with registration date, legal form and address, and a table of legal representatives with name, date and place of birth. We use the snapshot of 2 September 2026, key persons on name, birth date and birth place, hashed, and keep administrators and legal represen-

tatives. The register lists administrators as of the snapshot date, not shareholders and not historical administrators, so common administration is our proxy for common ownership; the open data carries a representatives table only in snapshots from November 2025 onwards, so administrators at the time of the 2018–2023 observations cannot be recovered, and a firm whose administrator changed since is mismeasured. Eighty-five percent of the limited-liability companies in the analysis sample have exactly one administrator, for whom the proxy is close; company names carry no usable marker of sole ownership. Nearly all statement filers, 96 to 99 percent by year, link to at least one administrator, and the linked share is the same across the position groups compared below. Four percent of administrator records lack a birth date, so their key is the name alone; results are reported with and without them. Names, birth dates and addresses are used only to build hashed keys and do not appear in any output or replication file.

Exchange rates. BNR reference rates at the last quotation of each year.

4 Empirical strategy

For each year t the notch location is $L_t = T_t e_{t-1}$, where T_t is the euro ceiling for revenue in year t and e_{t-1} the BNR rate at the close of year $t - 1$. Let F_t be the distribution of revenue in year t and $q_t = F_t(L_t)$ the quantile of the notch. For every other year c let $L_c^* = F_c^{-1}(q_t)$ be the revenue at the same quantile. Year c is an admissible control if no known notch of year c , the micro ceiling, an enacted or announced future ceiling, the €60,000 rate boundary, or the VAT threshold, lies within 25 percent of L_c^* . Bins are relative to the notch, 1 percent of the location wide, in a window of ± 20 percent, and shares $s_{t,k}$ are bin counts divided by the count in the window. The counterfactual share is the mean of $s_{c,k}$ over admissible controls. Excess mass is

$$b_t = \frac{\sum_{k \in B} (s_{t,k} - \bar{s}_{c,k})}{\frac{1}{|B|} \sum_{k \in B} \bar{s}_{c,k}}, \quad (1)$$

with B the five bins immediately below the notch, and the missing mass m_t is the same sum over the five bins above, negative when the treated density falls short of the control there. The identifying assumption is that the shape of the revenue density around a given quantile is the same across years up to the effect of the notch. Quantile matching makes the counterfactual invariant to inflation and to real growth of the size distribution, but not to changes in its shape, and because shares are normalised within the window the excess and missing masses must offset within it, so m_t is not independent evidence. We report a multinomial bootstrap, a block bootstrap that resamples control years, and, as the most relevant measure of uncertainty, the distribution of b at 77 placebo locations at 0.45 to 2.5 times each year’s ceiling that are themselves at least 25 percent from any

notch. Because b is expressed per bin it scales with bin width; with 1 percent bins b is approximately the width of the bunching region in percent of the ceiling under the usual approximation. We also report the excess number of firms below the notch, which does not depend on bin width, its range over regions of 3, 5 and 8 percent, and the excess as a share of all filers in the year, B/N_t . The admissibility rule is applied to the matched location rather than the window edges, so a control window can contain the region above another year’s notch; this affects only bins above the notch. Appendix Table 16 lists, for every notch and control year, the matched revenue and its distance to the nearest notch in that year, so the rule can be checked; the closest admissible cases are 26 to 34 percent away, all for the 2014–2016 notches against the VAT thresholds.

5 Results

5.1 The spike tracks the exchange rate

Within 2018–2022 the euro ceiling was constant at €1,000,000 but its lei value rose from 4.66 to 4.95 million lei as the leu depreciated. Figure 1 shows the revenue distribution in lei with 10,000-lei bins. The spike moves with the ceiling each year. Table 5 locates the modal 5,000-lei bin: in every year its centre lies 1,400 to 5,600 lei below the ceiling at the previous year-end rate, while the ceiling at the current year-end rate is 80,000 to 115,000 lei further right in 2019 to 2021 and within 5,000 lei in 2018 and 2022, when the leu barely moved. Firms target the in-year rule, which is also the lower of the two lei values, so the two explanations coincide. Nobody rounds revenue to 4,779,300 lei; a spike at that value that moves to 4,869,400 lei the next year is not round-number reporting or an accounting convention.

5.2 The spike follows the ceiling

Figure 2 shows a spike in the bin immediately below the ceiling in every year. It is at €65,000 in 2014 and 2015, moves to €100,000 in 2016 and €500,000 in 2017, sits

Table 5: Location of the revenue spike in lei, 2018–2022

Year	€1M, prev. rate (lei)	Modal 5,000-lei bin	Firms	Bin centre – threshold	Current – prev. rate
2018	4,659,700	4,655,000-4,660,000	59	-2,200	+4,200
2019	4,663,900	4,660,000-4,665,000	73	-1,400	+115,400
2020	4,779,300	4,775,000-4,780,000	95	-1,800	+90,100
2021	4,869,400	4,865,000-4,870,000	105	-1,900	+78,700
2022	4,948,100	4,940,000-4,945,000	101	-5,600	-700

Modal 5,000-lei bin within 60,000 lei of either candidate threshold. The last column gives the lei value of €1M at the current year-end rate minus its value at the previous year-end rate (the dotted line in Figure 1). At 1,000-lei resolution the modal bins hold only 16 to 31 firms and are noisier; they lie 2,000 to 5,000 lei below the previous-rate threshold in 2020–2022 and further below in 2018–2019.

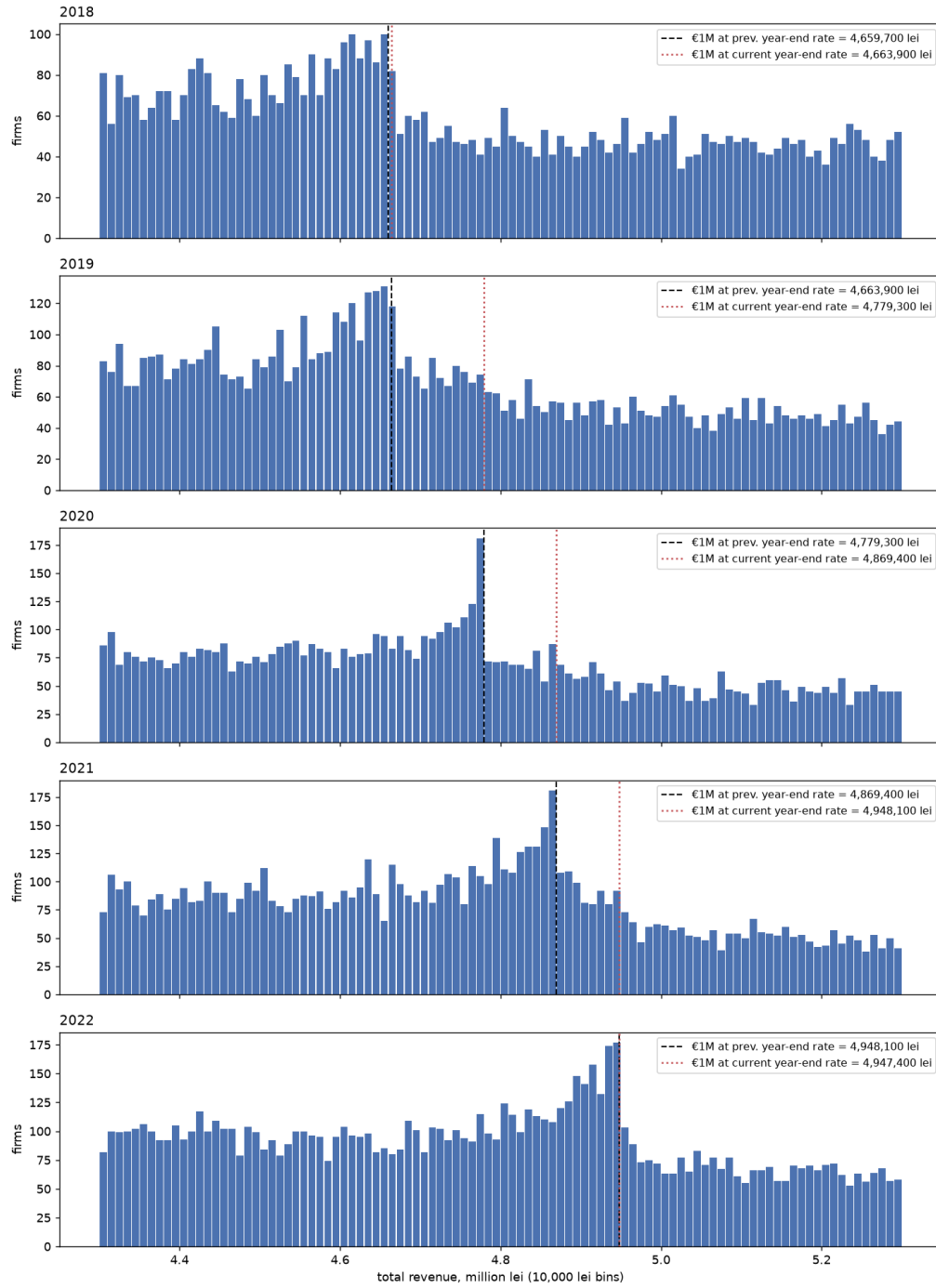


Figure 1: Total revenue in lei around €1,000,000, 2018–2022, 10,000-lei bins. Dashed: ceiling at the previous year-end rate; dotted: at the current year-end rate.

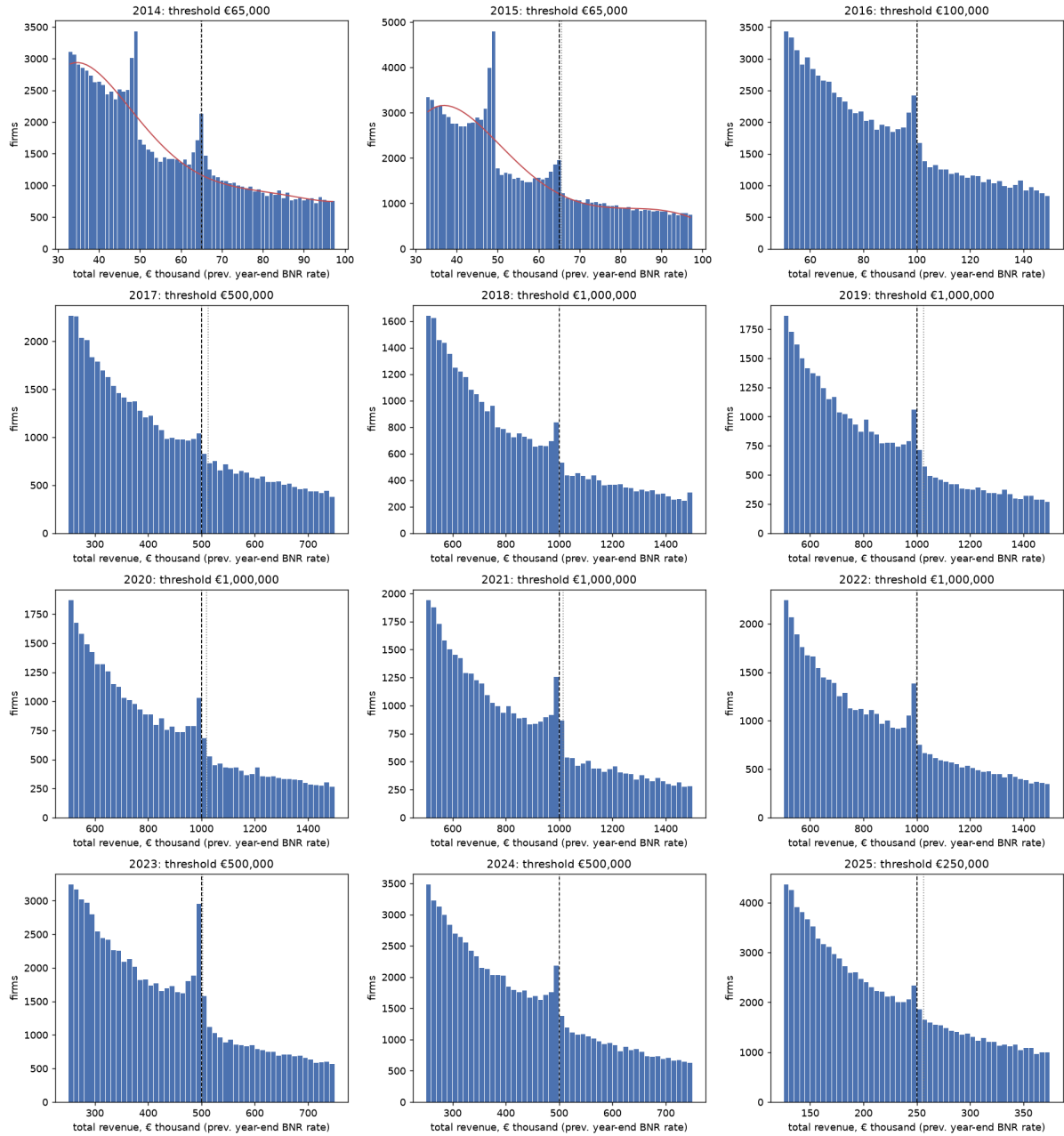


Figure 2: Total revenue around the micro-enterprise ceiling, 2014–2025, converted at the BNR rate at the close of the previous year. Dashed: ceiling; dotted: ceiling at the current year-end rate. In the 2014 and 2015 panels the red line is a polynomial counterfactual of degree 5 excluding three bins on each side, shown to make the point that it passes through the VAT spike, which lies inside its fitting window.

at €1,000,000 from 2018 to 2022, returns to €500,000 in 2023 and 2024 and moves to €250,000 in 2025. In 2014 and 2015 a second, taller spike is visible at about €49,000, which is 220,000 lei, the VAT threshold of the time. In 2014 the spike straddles the computed notch rather than sitting below it, consistent with the whole-year re-taxation rule then in force and with the base issue noted above.

5.3 Difference-in-bunching estimates

Table 6 reports the estimates and Figure 3 the densities for four representative notches; Appendix Figure 5 shows all fourteen. In every year the treated density exceeds the same-quantile control density in the bins just below the notch, and the control density is smooth through it. Excess mass in the 5 percent region below the notch ranges from 1.9 percent of firms in the window, at the €500,000 ceiling of 2017 and the €250,000 ceiling of 2025, to 7.6 percent at the €500,000 ceiling of 2023. At the 77 placebo locations b has mean 0.01, standard deviation 0.15 and a maximum absolute value of 0.55; every notch estimate exceeds the largest placebo, most by a factor of two or more, but the 2017 and 2025 in-year estimates, 0.76 and 0.73, are only one and a half placebo standard deviations above that extreme and should be read as the weakest in the set. For 2025 two things dilute the notch: the enacted €100,000 ceiling for 2026 competed for the same firms' attention, and at the 3 percent rate that applied above €60,000 the break-even margin is 18.75 percent, above the 13.5 percent median margin of that year's bunchers, so crossing was a tax increase for fewer firms.

As a share of all filers the excess is 0.06 to 0.30 percent of firms (Table 7). This is the number to carry into the policy discussion. The threshold response is large relative to the local density and negligible relative to the economy: the revenue at stake in the regime comes from the hundreds of thousands of firms taxed at 1 to 3 percent of revenue well below the ceiling, not from the few thousand that manage their revenue to stay under it. Table 7 shows that a block bootstrap resampling control years gives standard errors of 0.07 to 0.16, in line with the placebo spread, that dropping any single control year moves b by less than 0.1, and that narrowing the window lowers b and widening it raises it, because the normalisation is within the window, while the excess counts are unaffected. Appendix Table 17 adds that correcting the control quantile for the bunchers themselves changes b by at most 0.1, that control years before and after the treated year give similar estimates, and that restricting to firms present in the adjacent years lowers b by 0.05 to 0.2. The excess counts vary across the 3, 5 and 8 percent region definitions by between 6 and 50 percent, most at the two extra notches where the bunching region is wide. Excluding hospitality firms, which faced no revenue ceiling in 2023, raises the 2023 estimate from 2.99 to 3.14, and hospitality firms alone show 0.58.

Three patterns stand out. First, excess mass at the €1,000,000 ceiling grows from 3.5 percent of the window in 2018 to 5.5 percent in 2021, consistent with the gradual build-up of bunching found by Kleven and Waseem (2013). Second, 2023 is the largest response, at 7.6 percent, falling to 4.1 percent in 2024 at the same ceiling. Two things changed in 2023: the ceiling halved, so the density of firms at the new notch was high, and exit from the regime became permanent, so the cost of crossing rose. A natural conjecture is that firms displaced from above pulled revenue down to requalify, and 29.6 percent of the 2023

bunchers had indeed been above €500,000 in 2022; but the same is true at that location in years when it was not a threshold, since firms just below any point are disproportionately firms that were above it the year before. The ratio to a comparison group further below is 1.78 in 2023 against 1.55 to 1.96 in placebo years (Appendix Table 18), so the inflow from above is mean reversion. Newly created firms account for 11 percent of the 2023 excess, against 7 percent in 2021 and 13 percent in 2024, so entry at the threshold is a minor part of it. Third, the two extra rows show a second notch operating. In 2022 the ceiling in force was €1,000,000, but from July to December 2022 firms expected 2023 status to depend on 2022 revenue being below €500,000; a spike of 2.0 percent of the window appears there, with none at that location in 2021, and the December deferral came too late to undo it. In 2025 the same occurs at €100,000, enacted in December 2024 for 2026 status: 3.7 percent of the window in excess, located at the lei value of €100,000 at the end-2024 rate, the last rate firms could observe, rather than at the end-2025 rate the law prescribes. The modal 1,000-lei bin is 496,000 to 497,000 lei against 497,410 lei for €100,000 at the end-2024 rate; a round-number spike would sit at 499,000 to 500,000, and does not.

For comparison, a polynomial counterfactual of the kind used by Chetty et al. (2011) gives excess-mass estimates of 0.4 to 2.0 in a baseline specification, and across 27 combinations of bin width, polynomial degree and excluded window they range from negative values to more than twice the baseline in most years. Part of this is mechanical, since b scales with bin width, and in 2014 and 2015 the VAT spike lies inside the fitting window; but even holding bin width fixed the range across degrees and exclusion windows exceeds the bootstrap standard error several times over, the concern raised by Blomquist et al. (2021). The difference-in-bunching estimates involve no functional form; their choices, the window, the bin width and the exclusion rule for control years, are transparent, and the placebo distribution is the honest measure of their uncertainty.

Table 6: Difference-in-bunching estimates

Notch	Location (M lei)	Firms in window	b (SE)	m	Excess firms	% of window	Range 3–8%
2014: in-year, €65,000	0.292	33,859	1.15 (0.08)	0.77	986	2.9	932–931
2015: in-year, €65,000	0.291	34,896	1.71 (0.09)	-0.39	1,526	4.4	1,232–1,674
2016: in-year, €100,000	0.452	32,891	1.56 (0.08)	-0.25	1,321	4.0	985–1,493
2017: in-year, €500,000	2.271	17,172	0.76 (0.11)	-0.26	332	1.9	262–370
2018: in-year, €1,000,000	4.660	11,424	1.41 (0.14)	-0.59	406	3.5	341–467
2019: in-year, €1,000,000	4.664	13,105	1.67 (0.13)	0.05	558	4.3	494–605
2020: in-year, €1,000,000	4.779	12,693	1.75 (0.14)	-0.01	576	4.5	473–655
2021: in-year, €1,000,000	4.869	14,394	2.15 (0.13)	-0.08	793	5.5	670–916
2022: in-year, €1,000,000	4.948	16,448	1.92 (0.13)	-0.45	808	4.9	711–829
2023: in-year, €500,000	2.474	28,379	2.99 (0.11)	-0.14	2,151	7.6	1,850–2,388
2024: in-year, €500,000	2.487	28,619	1.62 (0.10)	-0.28	1,179	4.1	945–1,339
2025: in-year, €250,000	1.244	37,048	0.73 (0.08)	-0.06	690	1.9	588–697
2022: announced, €500,000	2.474	24,241	0.80 (0.10)	-0.13	494	2.0	359–531
2025: year-end, €100,000	0.497	55,232	1.47 (0.07)	0.47	2,059	3.7	1,616–2,278

Location = EUR threshold $\times$ BNR rate at the close of the previous year (in-year notches; the 2025 year-end notch at the end-2024 rate). Window $\pm 20\%$, bins 1% of the location. Control years are listed in Appendix Table 16. b = excess mass in the five bins below the notch in units of counterfactual firms per bin; m = the same over the five bins above, negative meaning a deficit (a hole) and positive an excess; SE from 300 multinomial bootstrap draws. Excess firms = excess in the 5% region below the notch, with its range over 3% and 8% regions.

Table 7: Robustness of the difference-in-bunching estimates

Notch	b	Block-bootstrap SE	Window $\pm 10\%$	Window $\pm 30\%$	Leave-one-out range	B/N_t (%)
2014: in-year, €65,000	1.15	0.08	0.74	0.78	1.13–1.18	0.20
2015: in-year, €65,000	1.71	0.08	1.36	1.55	1.69–1.72	0.30
2016: in-year, €100,000	1.56	0.11	1.23	1.70	1.54–1.64	0.25
2017: in-year, €500,000	0.76	0.12	0.62	0.79	0.73–0.79	0.06
2018: in-year, €1,000,000	1.41	0.14	1.20	1.46	1.38–1.43	0.07
2019: in-year, €1,000,000	1.67	0.14	1.27	1.87	1.63–1.71	0.09
2020: in-year, €1,000,000	1.75	0.16	1.35	1.93	1.71–1.83	0.09
2021: in-year, €1,000,000	2.15	0.15	1.59	2.39	2.10–2.20	0.13
2022: in-year, €1,000,000	1.92	0.14	1.58	2.15	1.87–1.97	0.12
2023: in-year, €500,000	2.99	0.11	2.19	3.35	2.97–3.02	0.30
2024: in-year, €500,000	1.62	0.09	1.26	1.78	1.58–1.64	0.15
2025: in-year, €250,000	0.73	0.08	0.57	0.78	0.71–0.75	0.09
2022: announced, €500,000	0.80	0.10	0.60	0.89	0.78–0.82	0.07
2025: year-end, €100,000	1.47	0.07	1.02	1.11	1.44–1.50	0.28

Baseline: window $\pm 20\%$, all admissible control years. Block-bootstrap SE: control years resampled with replacement and treated bin counts resampled, 500 draws. Leave-one-out: range of b dropping one control year at a time. B/N_t : excess firms in the 5% region as a share of all filers with positive revenue in year t . Further checks in Appendix Table 17.

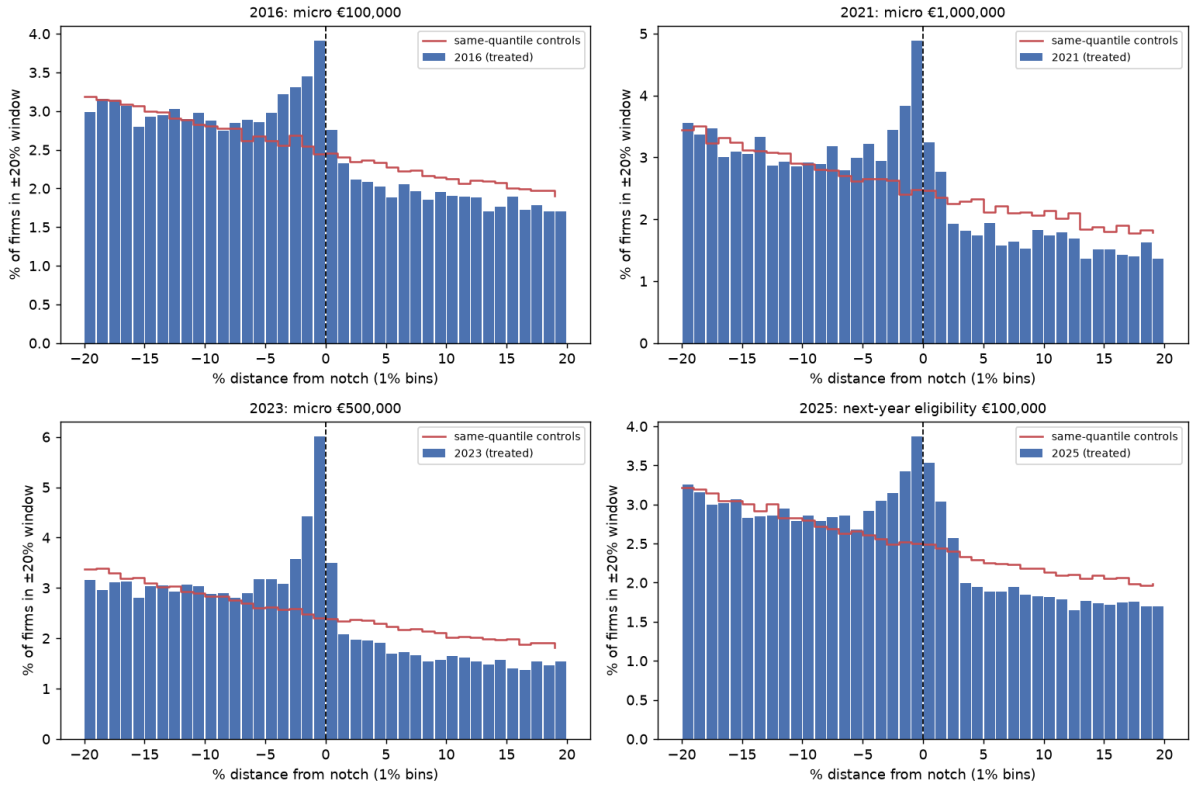


Figure 3: Treated-year density (bars) and same-quantile control density (line) around four notches, bins of 1 percent of the notch location. All fourteen notches in Appendix Figure 5.

5.4 Who bunches: margins and sectors

The notch is a tax increase only for firms whose margin exceeds the ratio of the two rates: 6.25 percent at a 1 percent revenue tax and 18.75 percent at 3 percent. Table 8 shows that firms in the 5 percent below the ceiling report median pre-tax margins of 11 to 20 percent, against 6 to 10 percent for firms just above and 7 to 12 percent for firms further below. Table 9 shows where they are. In 2023, excess mass is 12 to 15 percent of the window in professional, administrative, health and real-estate services and 10 percent in information and communication, against 5 percent in trade, 3.5 percent in transport and 1 percent in agriculture, and the bunchers in the service sectors report margins of 29 to 43 percent against 10 to 12 percent in trade and transport. The same ordering holds at the €250,000 ceiling of 2025, where information services lead at 5 percent and trade shows none. The sectors that bunch are the ones the 2023–2024 reforms singled out, and they are the ones where the regime saves the most tax per leu of revenue.

Table 8: Pre-tax profit margin by position relative to the threshold

Year	Bunchers (0–5% below)	Just above (0–10%)	10–25% below	Bunchers: margin > 6.25% (%)	Just above: margin > 6.25% (%)
2018	0.106	0.057	0.074	62	47
2019	0.117	0.070	0.090	65	53
2020	0.132	0.078	0.094	69	56
2021	0.160	0.097	0.122	74	63
2022	0.166	0.101	0.117	75	63
2023	0.200	0.098	0.124	76	60
2024	0.151	0.081	0.099	68	56
2025	0.135	0.093	0.101	64	58

Median of pre-tax profit over total revenue. At a 1% turnover tax and a 16% profit tax, the regime lowers the tax bill only for firms with margins above 6.25%; at 3% the break-even margin is 18.75%.

Table 9: Excess mass by sector, in percent of firms within 20% of the notch

CAEN section	2021 (€1M)	2023 (€500k)	2025 (€250k)	Firms in 2023 window	Median margin of 2023 bunchers
Professional services	–	14.8	4.6	2,068	0.39
Administrative services	–	13.3	3.4	1,089	0.29
Construction	7.4	12.6	3.1	4,078	0.25
Health	–	12.0	2.6	896	0.32
Real estate	–	11.2	3.5	908	0.43
Information and communication	–	10.1	5.1	1,124	0.34
Manufacturing	2.9	6.5	1.9	3,008	0.16
Trade	5.3	4.8	0.3	8,426	0.12
Transport	6.1	3.5	-0.2	2,428	0.10
Agriculture	0.5	1.0	1.0	1,653	0.08
Hospitality	–	0.9	0.4	1,571	0.14
Arts, other services	–	–	1.7	–	–

Difference-in-bunching within each CAEN section, matched at the section’s own quantile, same control years as the pooled estimate; sections with fewer than 800 firms in the window omitted. Hospitality faced no revenue ceiling in 2023.

5.5 Reported margins respond to the regime

Reported margins are not neutral to the regime: a micro-enterprise’s tax does not depend on its profit while a profit-tax firm’s does, so firms on profit tax have a reason to report lower margins that micro-enterprises lack. The ceiling changes let us test this directly. In 2018 the ceiling rose from €500,000 to €1,000,000, so every firm with 2017 revenue between those values was moved from profit tax to revenue tax without any action of its own, while firms above €1,000,000 stayed on profit tax; in 2023 the cut reversed the move. Wide bands, (€500,000, €1,000,000] against (€1,000,000, €2,000,000], compare firms that differ twofold in size and produce mean-reversion drift of up to 1.5 points in a placebo design (Appendix Table 19), which motivates the narrowed bands of Table 10: 20 percent either side of the cut-off, assigned on two-year average revenue, with placebos built at the same calendar time, the same bands at cut-offs of €1,500,000 and €2,000,000 in the same base years, where no regime changed. With narrow bands the raw pre-period differences are within 0.2 points of zero, and the contemporaneous placebos are within 0.5 points of zero before and after, so the correction is small and does not create a pre-trend. Entry into the revenue-tax regime in 2018 raises the treated group’s median margin by 0.7 points in the first year (standard error 0.2), from a base of about 6 percent; net of the placebos, 0.6 (0.3). Exit in 2023 lowers it by 1.0, 1.6 and 2.0 points over three years; net of the placebos, -0.8 (0.3), -1.5 (0.4) and -1.9 (0.3). Both effects have the predicted sign. The exit effect is the robust one, between 2.5 and 5.5 standard errors from zero at every horizon; the entry effect is significant only at the first horizon, 0.62 with a standard error of 0.26, and imprecise after, at 0.3 (0.3) and 0.5 (0.3). The asymmetry is itself sensible: cost misreporting is easier to start under a profit tax than to unwind on leaving it, and the exit effect builds over years, which is consistent with reporting adjusting gradually. Reported margins move by 10 to 25 percent of their level when a firm changes regime without changing anything else.

Table 10: Reported margins around regime switches: narrowed bands with contemporaneous placebos

Year	Horizon	DiD, median margin, pp (SE)	Placebo at €1.5M	Placebo at €2M	Corrected, pp (SE)
<i>2018 entry: bands on 2016–2017 average revenue, cut-off €1M; placebos at €1.5M and €2M</i>					
2015	-2	+0.09 (0.20)	+0.25	-0.06	-0.01 (0.28)
2016	-1	+0.16 (0.20)	+0.46	-0.04	-0.05 (0.27)
2017	+0	0.00	+0.00	+0.00	0.00
2018	+1	+0.71 (0.21)	+0.28	-0.11	+0.62 (0.26)
2019	+2	+0.51 (0.23)	+0.44	+0.03	+0.27 (0.29)
2020	+3	+0.48 (0.23)	+0.14	-0.08	+0.45 (0.28)
<i>2023 exit: bands on 2021–2022 average revenue, cut-off €1M; placebos at €1.5M and €2M</i>					
2020	-2	-0.06 (0.32)	+0.49	+0.04	-0.33 (0.41)
2021	-1	+0.07 (0.27)	+0.72	+0.21	-0.39 (0.32)
2022	+0	0.00	+0.00	+0.00	0.00
2023	+1	-0.95 (0.26)	+0.09	-0.36	-0.81 (0.33)
2024	+2	-1.62 (0.33)	-0.07	-0.20	-1.48 (0.38)
2025	+3	-1.98 (0.27)	-0.30	+0.06	-1.86 (0.34)

Balanced panels. Treated band: two-year average revenue in $(0.8, 1.0] \times$ the cut-off; control band: $(1.0, 1.2] \times$ the cut-off; DiD = change in the treated band’s median pre-tax margin minus the change in the control band’s, relative to the base year. Placebos: the same construction at cut-offs of €1.5M and €2M in the same base year, where no regime changed. Corrected = DiD minus the mean of the two placebos. SE from 200 bootstrap draws over firms. Firms: 5,353/3,760 treated/control in 2018, 7,556/4,766 in 2023; placebo designs 2,200–4,800 per band.

5.6 What the estimates do not identify

The standard route from excess mass to an elasticity runs through the width of the bunching region, found where the cumulative deficit above the notch offsets the excess below. In our data that point is not reached before the window edge in any year: the missing mass is spread over the whole window above the notch rather than concentrated next to it, which the within-window normalisation makes hard to distinguish from a level shift. Two further obstacles are specific to this notch. The tax change on crossing depends on the firm’s margin, since the base switches from revenue to profit, and at the 3 percent rate in force in 2014 to 2016 and 2024 to 2025 the median buncher’s reported margin lies below the 18.75 percent at which the regime saves tax, so a representative-firm elasticity has no clear meaning. And reported margins respond to the regime, as the previous subsection shows. We therefore report excess mass and excess shares. For completeness, Appendix Table 22 reports the Kleven–Waseem reduced-form elasticity under both proxies for the bunching width: read from b it is 0.003 to 0.02 in the 1 percent years and undefined in the 3 percent years, where the notch is not a tax increase at the median reported margin; read from the hole it is 1 to 5. Two proxies that differ by three orders of magnitude are the demonstration that the object is not identified here.

6 Other margins

6.1 The VAT threshold and the rate boundary

The VAT registration threshold moved from 300,000 to 395,000 lei on 1 September 2025, and the 2025 distribution of turnover shows the spike at 300,000 lei gone and a spike at 395,000 lei in its place, with polynomial excess mass of 3.0 against 0.6 at that location in 2024 (Table 11). Firms adjusted full-year turnover to a threshold that changed with four months of the year remaining, although part of the spike is mechanical, since the ordinance let firms that crossed 300,000 lei in August 2025 stay unregistered up to the new threshold. That relocation is our identification at the VAT threshold, because the difference-in-bunching estimator is not available there: a VAT notch exists in every year within 25 percent of the same-quantile location, so no year is an admissible control. The polynomial estimates in Table 11 and Appendix Figure 6 show excess mass between 2.8 and 6.6 in every year from 2014 to 2024 at bin widths of 0.7 to 0.9 percent of the threshold, against 0.4 to 2.0 for the micro-enterprise notch at bin widths of 1.5 to 2 percent; the VAT response is the larger even after allowing for the difference in bin width, and the spike moved from 220,000 to 300,000 lei when the threshold changed in April 2018. The legal base is the turnover of Article 310(2), which differs from net turnover by exempt and ancillary items; net turnover is blank for a fifth of micro-entity forms but for under 1 percent of firms near the threshold.

The move of the VAT threshold also uncovers the €60,000 boundary between the 1 and 3 percent rates, a notch of two points on the whole revenue base that exists from 2024. In 2024 the boundary and the VAT threshold lie 1,500 lei apart and the combined spike has polynomial excess mass of 4.9. In 2025, with the VAT threshold at 395,000 lei, a spike remains at 296,000 to 298,000 lei, the lei value of €60,000 at the end-2024 rate, with excess mass of 0.9 (Table 12, Figure 4): about a fifth of the combined 2024 spike, so most of the earlier spike was the VAT notch. Firms do respond to the rate boundary, but weakly relative to the exemption thresholds. Two points on the whole base is a large notch, and the weak response has a plausible cause: the activity list already puts software, legal and medical firms on 3 percent regardless of revenue, and those are the high-margin firms for which two points of revenue matter most, so the boundary binds mainly for firms with less at stake.

Table 11: Bunching of net turnover at the VAT registration threshold

Year	VAT threshold (lei)	Firms in window	b (polynomial)
2014	220,000	83,751	2.84
2015	220,000	91,256	4.21
2016	220,000	97,622	4.76
2017	220,000	105,826	5.39
2018	300,000	102,168	3.64
2019	300,000	113,716	5.09
2020	300,000	111,913	4.90
2021	300,000	117,918	5.78
2022	300,000	135,830	6.61
2023	300,000	145,471	6.62
2024	300,000	157,977	6.09
2025	300,000	157,676	0.79
2025	395,000	147,085	3.03

Polynomial counterfactual (degree 5, three bins excluded on each side), 2,000-lei bins, window $\pm 40\%$. The threshold rose to 395,000 lei on 1 September 2025; the two 2025 rows test the old and the new location (at 395,000 lei the 2024 placebo gives 0.62).

Table 12: The 1%/3% rate boundary at €60,000 and the VAT threshold, 2023–2025

Year	€60,000 in lei	VAT threshold (lei)	b at €60,000	b at 300,000	Modal 2,000-lei bin, 250–450k	Firms in modal / median bin
2023	296,844	300,000	3.17	5.89	298,000-300,000	5,215 / 742
2024	298,476	300,000	4.88	5.53	298,000-300,000	5,121 / 801
2025	298,446	300,000 (to Aug. 2025)	0.94	0.92	296,000-298,000	1,910 / 1,101

Polynomial excess mass, 1,500-lei bins, window $\pm 30\%$. In 2023 no rate boundary existed; in 2024 the boundary and the VAT threshold lie 1,500 lei apart; in 2025 the VAT threshold moved to 395,000 lei on 1 September.

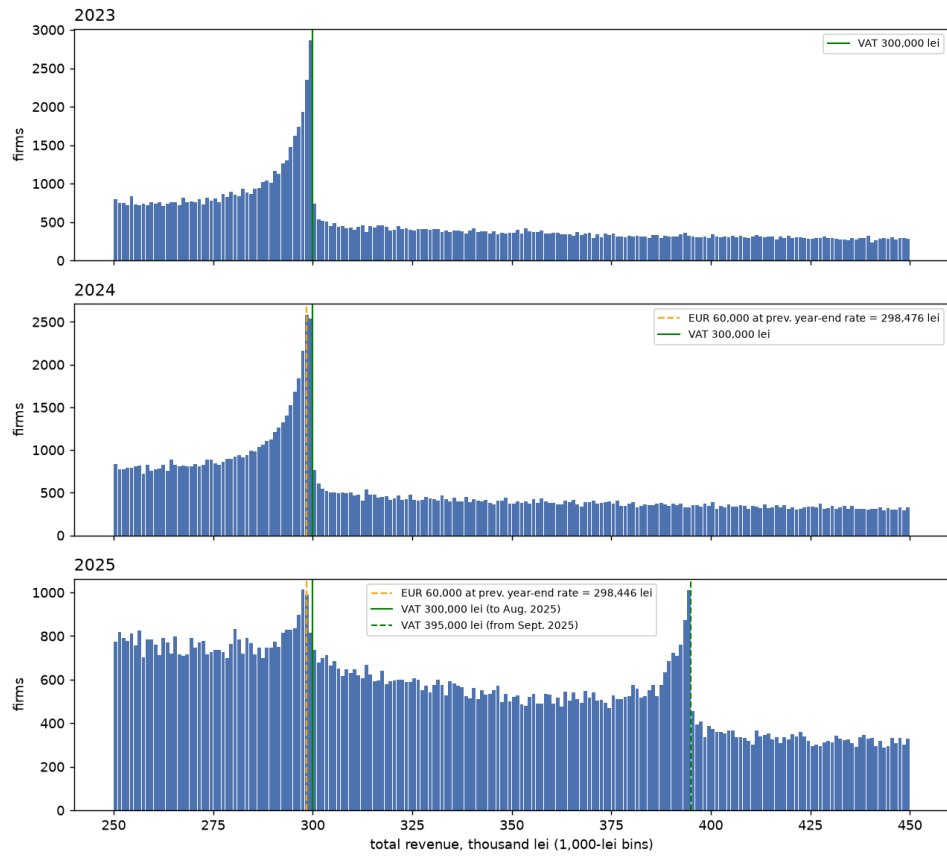


Figure 4: Total revenue from 250,000 to 450,000 lei, 2023–2025, 1,000-lei bins, with the €60,000 rate boundary and the VAT thresholds.

6.2 The employee condition

From 1 January 2023 a micro-enterprise must have at least one employee, tested at 31 December 2022; the condition was enacted in July 2022. Table 13 shows the distribution of headcount among firms with revenue below €500,000: the share with exactly one employee rose from 31.9 percent in 2021 to 37.6 percent in 2022 and 40.7 percent in 2023, while the share with five or more fell. Table 14 follows zero-employee firms into the next year. In the 2021 to 2022 transition 22.3 percent of them moved to exactly one employee, against 11.1 to 11.3 percent in the two preceding transitions, or 20.3 percent when the sample is restricted to firms present since 2020 to hold filing coverage fixed; among zero-employee firms above €1,500,000, which the rule did not touch, the rate stayed at 5 to 6 percent throughout. This is about 20,000 additional one-employee firms. Of micro-enterprises that still had no employees at the end of 2022, 82 percent carried the profit-tax registration flag in 2023, and 15 percent had one employee and 3 percent two or more by the end of 2023. The hiring response came in the enactment year; firms that had not hired by the deadline left the regime rather than hire afterwards. Among the 2022 hirers, 31 percent paid a blended rate that year, consistent with a hire during the year, and 12 percent the no-employee rate, consistent with a hire in the last quarter; the corresponding shares in the 2021 placebo are 14 and 19 percent. We read the response as compliance at the minimum cost rather than as job creation.

Table 13: Employee counts among firms with revenue below €500,000 (% of firms)

Year	0 employees	1	2	5+	Firms
2019	33.6	28.7	13.9	11.8	555,050
2020	33.4	30.6	13.4	11.2	575,855
2021	35.1	31.9	12.1	10.2	558,724
2022	31.1	37.6	11.9	9.1	616,503
2023	28.7	40.7	12.5	8.3	640,536
2024	30.6	40.8	12.2	7.3	686,605
2025	32.6	40.5	11.8	6.4	655,235

Average annual headcount as reported in the statements.

Table 14: Transitions of zero-employee firms (revenue below €500,000) into the next year, conditional on filing in both years (%)

Year t	Firms with 0 employees in t	1 employee in $t + 1$	2+ in $t + 1$	Still 0 in $t + 1$
2019	186,324	11.1	4.7	84.1
2020	192,284	11.3	3.5	85.2
2021	195,972	22.4	3.3	74.5
2022	191,971	14.8	3.0	82.3
2023	183,852	9.4	2.9	87.7
2024	210,272	8.1	2.2	89.6

The one-employee condition was enacted in July 2022 and tested at 31 December 2022. Filing coverage differs across years because some annual files are early uploads, so shares are conditional on a statement being present in $t + 1$.

6.3 Splitting

Table 15 compares limited-liability companies in the 5 percent below the ceiling with firms just above it, which are of the same size, and with firms further below and well above. Pooled over 2018–2023, 2.7 percent of bunchers have an administrator who registered another firm in the preceding eighteen months, against 2.0 percent of firms just above ($z = 4.3$, standard errors clustered by firm throughout) and 1.7 percent further below; 12.2 percent have another firm newly registered at their address, against 9.2 and 8.7 percent ($z = 8.6$). Dropping administrators without a birth date, whose key may merge namesakes, gives 2.4 against 1.7 percent, the same ratio. The share whose administrator runs any other firm, regardless of date, differs little between bunchers and firms just above ($z = 2.6$) and is similar for bunchers and firms well above the ceiling, so the timing measures rather than the level carry the information.

Bunchers are also younger than firms just above, 13 percent under three years old against 8 percent, and a recently registered sibling is more likely for a young firm. Comparing within age bands (Appendix Table 20), bunchers exceed the just-above group in every band, and standardising to the bunchers’ age distribution the premium is 23 percent on the administrator measure and 20 percent on the address measure. At the small ceilings of 2015–2017 the administrator measure shows no difference ($z = 1.5$, Table 15 note).

The 2024 rules provide two checks. From 2023 a person could hold over 25 percent in at most three micro-enterprises, and from 2024 in one, while the ceiling began to be tested on the combined revenue of linked enterprises. If the premium reflects splitting to stay under the ceiling, it should fall once splitting no longer helps, and combined revenue should itself start to bunch. The first prediction holds on the administrator measure: bunchers and firms just above are at 1.4 percent each in 2024, against 2.0 and 1.6 in 2023; a gap remains on the address measure. The second does not: the combined revenue of firms sharing an administrator shows no excess mass at €500,000 in 2024, and firms

Table 15: Sibling-firm indicators by position relative to the threshold, SRLs, pooled 2018–2023

Measure	Bunchers	Just above	10–25% below	25–100% above	z
Administrator registered another firm, prior 18 months	2.71	1.98	1.72	1.91	4.3
New firm at the same address, prior 18 months	12.23	9.18	8.72	8.37	8.6
Administrator runs another firm, any date	17.09	15.88	15.16	17.46	2.6

Pooled shares, %. z tests bunchers against firms just above the threshold with standard errors clustered by firm. At the small ceilings of 2015–2017 the corresponding shares for bunchers and firms just above are 1.59 versus 1.37 ($z = 1.5$) on the administrator measure, 7.17 versus 5.34 ($z = 6.3$) on the address measure and 12.42 versus 11.38 ($z = 2.7$) on the any-date measure. Standardising the just-above group to the bunchers’ age distribution gives 2.70 versus 2.20 on the administrator measure and 12.26 versus 10.23 on the address measure (Appendix Table 20). Administrators as recorded in the ONRC snapshot of 2 September 2026; persons administering more than 50 firms and addresses hosting more than 20 firms excluded. Bunchers: 0–5% below the threshold; just above: 0–10% above.

inside such groups bunch individually at the same rate as single firms (Appendix Table 21). Cohorts of firms that lost the regime when the ceiling fell registered new firms at the same rate relative to the band below as placebo cohorts, so whatever splitting there is serves to stay under the line rather than to react to losing the regime.

Either the aggregation rule was not enforced or salient in its first year, or common administration is too weak a proxy for the ownership links the rule is written on; the evidence does not distinguish these, and it means the splitting result should be read as a modest, robust correlation rather than an established mechanism. The register lists administrators as of 2026 rather than shareholders at the time, and the address measure will catch shared accountants’ premises despite the exclusion of addresses hosting more than twenty firms.

7 Conclusion

Romania’s micro-enterprise ceiling moved six times in twelve years and the distribution of reported firm revenue moved with it, to within a few thousand lei of the ceiling’s value at the exchange rate written into the law. The variation identifies bunching without a polynomial, under a stated and tested assumption, and at locations with no notch the estimator gives excess mass centred on zero, with a spread that every notch estimate exceeds. Excess mass is 2 to 8 percent of firms near the ceiling and 0.06 to 0.30 percent of all firms: the threshold is where the regime is visible, not where its cost lies. The channels are consistent with a tax-saving motive concentrated in high-margin services: bunching is two to three times stronger in professional and information services than in trade, reported margins rise by about 0.7 points when firms are moved onto the revenue

tax and fall by 1 to 2 points when they are moved off it, firms took on a single employee in the year the employee condition was enacted or left the regime, and firms at the ceiling are somewhat more likely than equally sized and aged firms above it to have spun off a sibling firm, a premium that faded when linked enterprises began to be aggregated, even though combined revenue does not bunch. A static revenue simulation of the kind used in recent Romanian policy debate (Ban and Buciu, 2023) assumes none of these responses; the estimates here are the behavioural inputs such simulations lack, and their aggregate size says that the ceiling itself is a second-order concern. Converting the excess mass into an elasticity requires a model in which the tax base changes at the threshold, margins are heterogeneous and reported margins respond to the regime; the missing mass above the notch is too diffuse in these data to support the standard shortcut.

References

- Almunia, M. and D. López-Rodríguez (2018). Under the radar: The effects of monitoring firms on tax compliance. *American Economic Journal: Economic Policy* 10(1), 1–38.
- Asatryan, Z. and A. Peichl (2016). Responses of firms to tax, administrative and accounting rules: Evidence from Armenia. ZEW Discussion Paper 16-065.
- Bachas, P. and M. Soto (2021). Corporate taxation under weak enforcement. *American Economic Journal: Economic Policy* 13(4), 36–71. doi:10.1257/pol.20180564.
- Ban, C. and P. Buciu (2023). *Reforma fiscală: propuneri pe termen scurt*. Bucharest: Friedrich-Ebert-Stiftung.
- Best, M. and H. Kleven (2018). Housing market responses to transaction taxes: Evidence from notches and stimulus in the UK. *Review of Economic Studies* 85(1), 157–193. doi:10.1093/restud/rdx032.
- Blomquist, S., W. Newey, A. Kumar and C.-Y. Liang (2021). On bunching and identification of the taxable income elasticity. *Journal of Political Economy* 129(8), 2320–2343.
- Boonzaaier, W., J. Harju, T. Matikka and J. Pirttilä (2019). How do small firms respond to tax schedule discontinuities? Evidence from South African tax registers. *International Tax and Public Finance* 26(5), 1104–1136. doi:10.1007/s10797-019-09550-z.
- Chetty, R., J. Friedman, T. Olsen and L. Pistaferri (2011). Adjustment costs, firm responses, and micro vs. macro labor supply elasticities: Evidence from Danish tax records. *Quarterly Journal of Economics* 126(2), 749–804.
- Consiliul Fiscal (2024). *Raport anual 2023*. Bucharest.

- Dharmapala, D., J. Slemrod and J. Wilson (2011). Tax policy and the missing middle: Optimal tax remittance with firm-level administrative costs. *Journal of Public Economics* 95(9–10), 1036–1047.
- Harju, J., T. Matikka and T. Rauhanen (2019). Compliance costs vs. tax incentives: Why do entrepreneurs respond to size-based regulations? *Journal of Public Economics* 173, 139–164.
- International Monetary Fund (2025). *Romania: Technical Assistance Report, High-Level Summary*. Washington, DC.
- Kanbur, R. and M. Keen (2014). Thresholds, informality, and partitions of compliance. *International Tax and Public Finance* 21(4), 536–559.
- Kleven, H. and M. Waseem (2013). Using notches to uncover optimization frictions and structural elasticities: Theory and evidence from Pakistan. *Quarterly Journal of Economics* 128(2), 669–723.
- Lazăr, S. (2024). Effective tax rates and firm size under turnover tax: Evidence from a natural experiment on SMEs. *Economics* 18(1), article 20220079. doi:10.1515/econ-2022-0079.
- Liu, L., B. Lockwood, M. Almunia and E. Tam (2021). VAT notches, voluntary registration, and bunching: Theory and UK evidence. *Review of Economics and Statistics* 103(1), 151–164.
- Marx, B. (2024). Dynamic bunching estimation with panel data. *Journal of Econometric Methods* 13(2), 225–249. doi:10.1515/jem-2022-0031.
- Sharma, R., J. Slemrod, M. Stimmelmayer, J. Wilson and P. Choi (2025). Optimal dual-regime business tax systems. NBER Working Paper 34418. doi:10.3386/w34418.

A Legal timeline

The regime’s parameters were changed by Ordinance 8/2013 (mandatory regime, €65,000, 3 percent, from 1 February 2013), Ordinance 102/2013 (consulting cap, 2014), Ordinance 50/2015 (€100,000 and employee-dependent rates, 2016), Ordinance 3/2017 (€500,000 from 1 February 2017), Ordinance 79/2017 (€1,000,000 from 2018, consulting cap abolished), Ordinance 25/2018 (capital and employee opt-out from April 2018), Ordinance 16/2022 of 15 July 2022 (€500,000, employee condition, three-firm rule, optional regime with permanent exit, from 2023) as amended by Law 370/2022 of 20 December 2022 (revenue test deferred to 2023 revenue), Law 296/2023 (€60,000 rate boundary and activity list, 2024), Ordinance 115/2023 (one-firm rule and linked-enterprise aggregation, 2024), Ordinance 156/2024 of 31 December 2024 (€250,000 for 2025, €100,000 for 2026, consulting cap abolished), Ordinance 89/2025 (flat 1 percent from 2026) and Ordinance 8/2026 (turnover base from 2026). Exchange-rate provisions: Article 47(1)(c) for the year-end test and Article 52 for the in-year test of Law 227/2015, and before 2016 Article 112⁶ of Law 571/2003 with its implementing norms; in every version the in-year test uses the BNR rate at the close of the previous financial year. VAT thresholds: Article 310 of Law 227/2015 as amended by Law 72/2018 (300,000 lei from 1 April 2018) and Ordinance 22/2025 (395,000 lei from 1 September 2025). A full table with Monitorul Oficial references accompanies the replication files. Policy assessments of the regime are in IMF (2025), Consiliul Fiscal (2024), Ban and Buciu (2023) and Lazăr (2024).

B Additional tables

Table 16 lists the admissible control years for each notch with the matched revenue and its distance to the nearest notch. Table 17 gives the robustness checks not shown in the body. Table 18 gives the origin of firms just below €500,000 in the notch year and in placebo years. Table 19 gives the wide-band event study that motivates the narrowed bands in the body. Tables 20 and 21 give the age-band comparison and the combined-revenue test for the splitting section. Table 22 gives the reduced-form elasticities discussed in Section 5.

Table 16: Admissible control years, matched locations and distance to the nearest notch

Notch	Control year: matched revenue L_c^* (distance to nearest notch in that year)
2014: in-year, €65,000	2016: 0.32M (32%); 2017: 0.33M (34%); 2022: 0.44M (31%); 2023: 0.46M (35%); 2024: 0.47M (36%)
2015: in-year, €65,000	2016: 0.30M (26%); 2017: 0.30M (27%); 2023: 0.43M (30%); 2024: 0.43M (31%)
2016: in-year, €100,000	2014: 0.42M (30%); 2015: 0.47M (38%); 2017: 0.48M (54%); 2018: 0.50M (40%); 2019: 0.52M (42%); 2020: 0.49M (39%); 2021: 0.57M (47%); 2022: 0.62M (52%); 2023: 0.66M (54%); 2024: 0.66M (54%)
2017: in-year, €500,000	2014: 2.09M (86%); 2015: 2.25M (87%); 2016: 2.25M (80%); 2018: 2.42M (88%); 2019: 2.50M (86%); 2020: 2.39M (88%); 2021: 2.80M (74%); 2025: 3.13M (60%)
2018: in-year, €1,000,000	2014: 4.19M (93%); 2015: 4.49M (94%); 2016: 4.44M (90%); 2017: 4.54M (50%); 2023: 5.83M (58%); 2024: 5.88M (58%); 2025: 6.26M (80%)
2019: in-year, €1,000,000	2014: 4.12M (93%); 2015: 4.41M (93%); 2016: 4.37M (90%); 2017: 4.46M (49%); 2023: 5.73M (57%); 2024: 5.79M (57%); 2025: 6.16M (80%)
2020: in-year, €1,000,000	2014: 4.42M (93%); 2015: 4.73M (94%); 2016: 4.69M (90%); 2017: 4.76M (52%); 2023: 6.15M (60%); 2024: 6.21M (60%); 2025: 6.60M (81%)
2021: in-year, €1,000,000	2014: 3.92M (93%); 2015: 4.19M (93%); 2016: 4.15M (89%); 2017: 4.24M (46%); 2023: 5.44M (54%); 2024: 5.48M (55%); 2025: 5.84M (79%)
2022: in-year, €1,000,000	2014: 3.56M (92%); 2015: 3.82M (92%); 2016: 3.79M (88%); 2017: 3.87M (41%); 2023: 4.94M (50%); 2024: 5.01M (50%); 2025: 5.32M (77%)
2023: in-year, €500,000	2014: 1.79M (84%); 2015: 1.94M (85%); 2016: 1.94M (77%); 2018: 2.09M (86%); 2019: 2.16M (86%); 2020: 2.07M (86%); 2021: 2.42M (88%); 2025: 2.70M (54%)
2024: in-year, €500,000	2014: 1.75M (83%); 2015: 1.91M (85%); 2016: 1.90M (76%); 2018: 2.05M (85%); 2019: 2.12M (86%); 2020: 2.03M (85%); 2021: 2.37M (87%); 2025: 2.65M (53%)
2025: in-year, €250,000	2014: 0.82M (64%); 2015: 0.90M (68%); 2016: 0.89M (49%); 2017: 0.92M (76%); 2018: 0.97M (69%); 2019: 1.01M (70%); 2020: 0.96M (69%); 2021: 1.11M (73%); 2022: 1.21M (75%); 2023: 1.25M (76%); 2024: 1.23M (76%)
2022: announced, €500,000	2014: 1.69M (83%); 2015: 1.84M (84%); 2016: 1.83M (75%); 2018: 1.97M (85%); 2019: 2.05M (85%); 2020: 1.96M (85%); 2021: 2.28M (87%); 2025: 2.56M (51%)
2025: year-end, €100,000	2016: 0.34M (31%); 2017: 0.35M (37%); 2021: 0.42M (29%); 2022: 0.46M (35%); 2023: 0.49M (39%); 2024: 0.50M (40%)

L_c^* = revenue at the notch's quantile in the control year. Distance = $|n/L_c^* - 1|$ for the nearest known notch n of that year (micro ceiling, future ceiling, €60,000 boundary or VAT threshold); admissible if above 25%. The closest cases are 26–34%.

Table 17: Further robustness checks of the difference-in-bunching estimates

Notch	b	Quantile-corrected	Pre-years only	Post-years only	Continuing firms	n continuing	Jackknife SE
2014: in-year, €65,000	1.15	1.14	–	1.15	1.15	31,696	0.04
2015: in-year, €65,000	1.71	1.70	–	1.71	1.71	30,382	0.02
2016: in-year, €100,000	1.56	1.56	1.30	1.63	1.45	29,322	0.08
2017: in-year, €500,000	0.76	0.77	0.62	0.85	0.72	16,321	0.05
2018: in-year, €1,000,000	1.41	1.36	1.44	1.37	1.35	11,010	0.04
2019: in-year, €1,000,000	1.67	1.75	1.64	1.70	1.58	12,513	0.06
2020: in-year, €1,000,000	1.75	1.87	1.64	1.90	1.67	11,535	0.09
2021: in-year, €1,000,000	2.15	2.19	2.23	2.04	2.08	13,668	0.08
2022: in-year, €1,000,000	1.92	1.97	1.82	2.05	1.77	14,756	0.09
2023: in-year, €500,000	2.99	2.99	3.02	2.82	2.80	26,100	0.05
2024: in-year, €500,000	1.62	1.65	1.64	1.52	1.58	24,967	0.04
2025: in-year, €250,000	0.73	0.72	0.73	–	0.61	35,470	0.04
2022: announced, €500,000	0.80	0.79	0.80	0.83	0.83	21,379	0.03
2025: year-end, €100,000	1.47	1.47	1.47	–	1.43	51,879	0.04

Quantile-corrected: control quantile shifted down by B/N_t . Pre/post: only control years before/after the treated year. Continuing: firms with positive revenue in $t - 1$, t and $t + 1$. Jackknife SE over control years.

Table 18: Origin of firms just below €500,000: share that was above €500,000 the year before

Year	Status	Firms 0–6% below	% above €500k in $t-1$	% in 120–200% band in $t-1$	Firms 14–20% below	% above in $t-1$	Ratio
2019	placebo	2,999	23.6	8.3	3,922	12.3	1.92
2020	placebo	2,895	40.7	17.5	3,817	25.0	1.63
2021	placebo	3,058	19.4	7.3	3,999	9.9	1.96
2023	notch year	6,646	29.6	15.1	5,166	16.6	1.78
2024	post-transition	5,659	21.7	7.3	5,403	14.0	1.55

Positions relative to €500,000 at the previous year-end rate in every year, whether or not it was a threshold. Ratio = bunchers' share from above divided by the comparison group's share.

Table 19: Reported pre-tax margins around regime switches induced by ceiling changes

Year	Micro share T / C (%)	Median margin T (%)	Median margin C (%)	DiD median (pp)	DiD mean (pp)	Median revenue T / C (k€)
<i>2018 increase: (500k, 1M] vs (1M, 2M] on 2017 revenue</i>						
2015	15 / 14	5.39	5.07	-0.32	-0.39	559 / 1,108
2016	14 / 14	5.80	5.41	-0.25	-0.43	628 / 1,232
2017	27 / 15	6.00	5.36	+0.00	+0.00	684 / 1,351
2018	72 / 18	6.37	4.93	+0.80	+0.85	728 / 1,438
2019	61 / 22	6.22	4.89	+0.69	+0.65	746 / 1,466
2020	58 / 26	6.07	5.06	+0.37	-0.08	675 / 1,342
<i>2023 cut: (500k, 1M] vs (1M, 2M] on 2022 revenue</i>						
2020	74 / 51	9.84	7.70	-0.70	-1.14	462 / 912
2021	75 / 48	11.87	9.36	-0.33	-0.51	576 / 1,090
2022	75 / 36	11.76	8.92	+0.00	+0.00	696 / 1,351
2023	55 / 24	8.40	6.69	-1.13	-1.49	676 / 1,358
2024	24 / 18	6.43	5.69	-2.10	-2.49	691 / 1,381
2025	18 / 17	5.80	5.35	-2.39	-3.22	686 / 1,366
<i>Placebo: same bands on 2019 revenue, no regime change</i>						
2017	46 / 24	7.35	5.66	-1.47	-1.60	533 / 1,099
2018	74 / 37	8.24	5.82	-0.74	-0.88	621 / 1,249
2019	71 / 26	8.95	5.79	+0.00	+0.00	692 / 1,364
2020	70 / 21	7.85	5.54	-0.85	-0.98	640 / 1,270
2021	66 / 29	9.43	6.63	-0.36	-1.17	718 / 1,421

Balanced panels of firms with positive revenue in every year shown. T = firms whose base-year revenue placed them in the band that changed regime; C = firms in the band above that did not. DiD = change in T minus change in C relative to the base year (2017, 2022, 2019). Margins are pre-tax profit over revenue, winsorised at ± 1 . The placebo shows the drift induced by selecting on base-year revenue alone.

Table 20: Sibling-firm indicators by firm age, bunchers versus firms just above the threshold, pooled 2018–2023 (%)

Firm age (years)	Bunchers, n	Just above, n	Admin. new firm: bunchers	just above	Same-address new firm: bunchers	just above
0-2	2,166	1,482	6.97	5.47	26.13	20.51
3-5	2,544	2,332	3.26	2.57	18.20	16.51
6-10	3,662	3,826	2.48	2.12	13.68	11.53
11+	8,158	10,187	1.50	1.25	6.09	4.96

Age = year minus year of registration in the trade register. Standardising the just-above group to the bunchers' age distribution gives 2.70 versus 2.20 (administrator measure, +23%) and 12.26 versus 10.23 (address measure, +20%).

Table 21: Bunching of combined revenue of firms sharing an administrator, and firm-level bunching inside groups

Year	Groups (2–5 SRLs, one administrator)	Groups in window	b , combined revenue at €500k	Excess groups	Firms in groups: % in 5% below own notch	Single firms: same
2021	13,563	970	0.11	3	19.5	18.1
2022	15,156	1,118	0.58	16	18.9	17.5
2023	15,860	1,304	0.90	29	21.5	20.1
2024	17,323	1,417	0.07	3	16.2	16.9
2025	18,044	1,335	0.62	20	15.8	14.5

From 2024 the ceiling is tested on the combined revenue of linked enterprises. Combined revenue of each administrator's firms, window $\pm 20\%$ around €500,000 at the previous year-end rate, controls = same-quantile density of combined revenue in 2019–2022 excluding years where the individual €1M notch is within 25%. Last two columns: share of firms within 20% of that year's individual notch that sit in the 5% just below it.

Table 22: Reduced-form elasticities under the Kleven–Waseem approximation, reported for completeness

Notch	τ	median buncher margin	Δt (pp)	$\Delta z^*/z^*$ from b (%)	e from b	hole width (%)	e from hole
2014: in-year, €65,000	3%	0.160	-0.44	1.15	undefined (dt<=0)	–	undefined (dt<=0)
2015: in-year, €65,000	3%	0.169	-0.30	1.71	undefined (dt<=0)	20	undefined (dt<=0)
2016: in-year, €100,000	3%	0.134	-0.86	1.56	undefined (dt<=0)	19	undefined (dt<=0)
2017: in-year, €500,000	1%	0.089	+0.42	0.76	0.007	20	4.67
2018: in-year, €1,000,000	1%	0.106	+0.70	1.41	0.014	18	2.3
2019: in-year, €1,000,000	1%	0.117	+0.87	1.67	0.016	18	1.84
2020: in-year, €1,000,000	1%	0.132	+1.11	1.75	0.014	19	1.61
2021: in-year, €1,000,000	1%	0.160	+1.56	2.15	0.015	19	1.15
2022: in-year, €1,000,000	1%	0.166	+1.66	1.92	0.011	20	1.2
2023: in-year, €500,000	1%	0.200	+2.20	2.99	0.02	–	hole not identified
2024: in-year, €500,000	3%	0.151	-0.58	1.62	undefined (dt<=0)	19	undefined (dt<=0)
2025: in-year, €250,000	3%	0.135	-0.84	0.73	undefined (dt<=0)	20	undefined (dt<=0)
2022: announced, €500,000	1%	0.140	+1.24	0.80	0.003	18	1.29
2025: year-end, €100,000	3%	0.234	+0.74	1.47	0.014	20	2.61

$\Delta t = 0.16 \times \text{margin} - \tau$ is the change in the average tax rate on revenue at the notch for a firm with the median reported margin of bunchers; $e = (\Delta z^*/z^*)^2/(2\Delta t/(1 - \tau))$. Two proxies for $\Delta z^*/z^*$: b with 1% bins, and the width above the notch at which the cumulative deficit offsets the excess (the hole), which is not reached before the window edge in most years. The proxies differ by three orders of magnitude and $\Delta t \leq 0$ in the 3% years, which is why the text does not rely on these numbers.

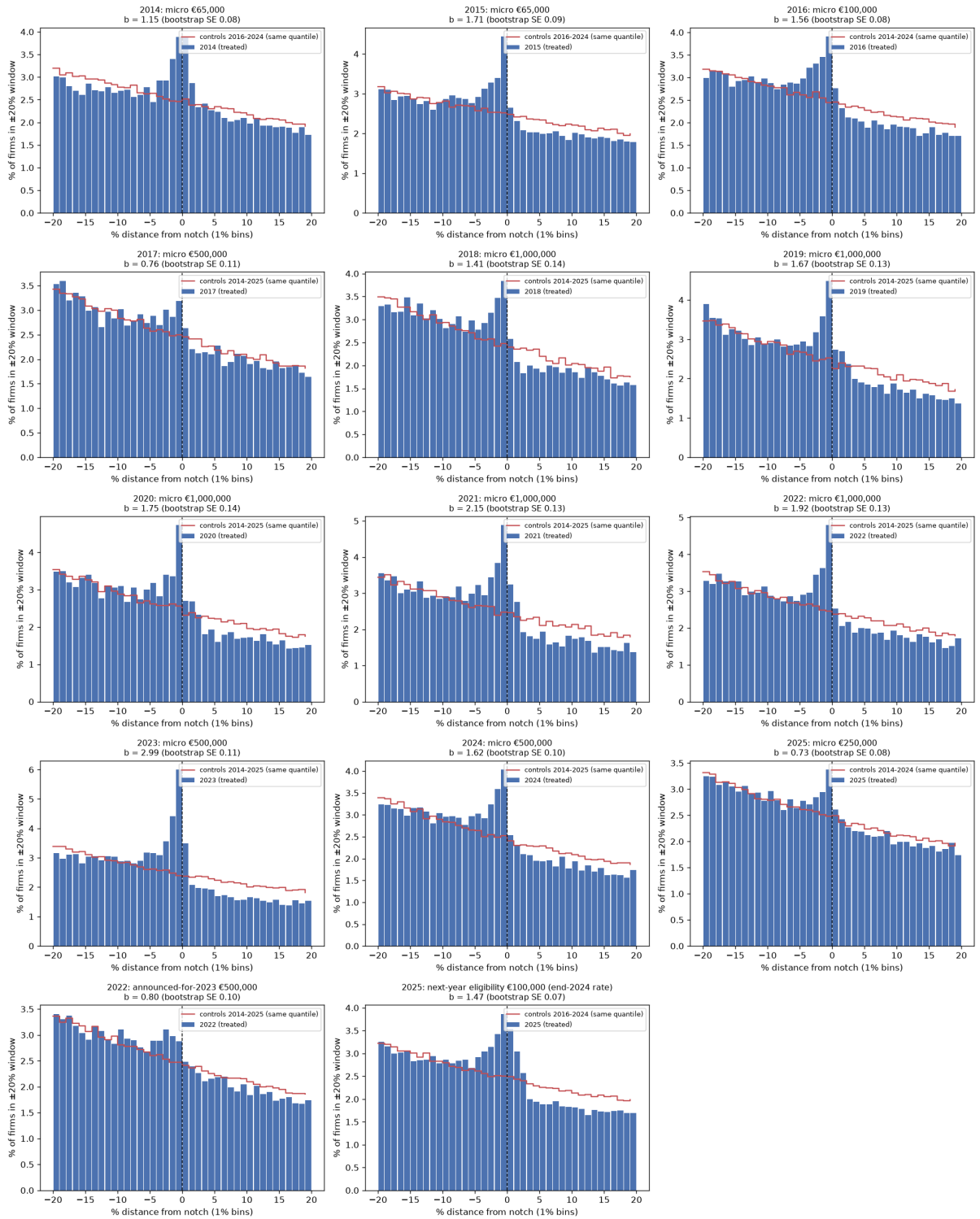


Figure 5: Treated-year density (bars) and same-quantile control density (line) around each of the fourteen notches, bins of 1 percent of the notch location.

C Additional figures

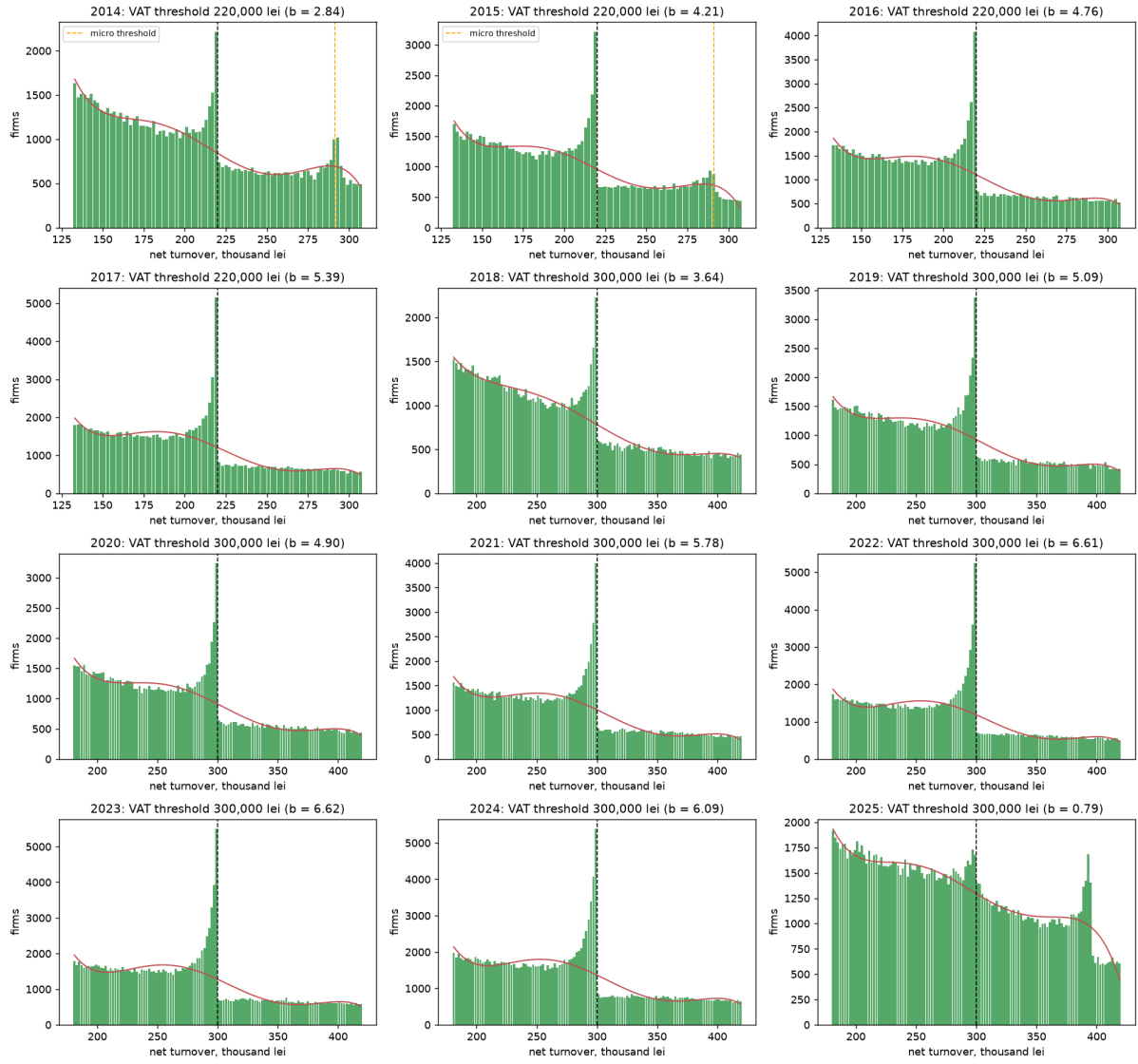


Figure 6: Net turnover around the VAT registration threshold, 2,000-lei bins, with a polynomial counterfactual.

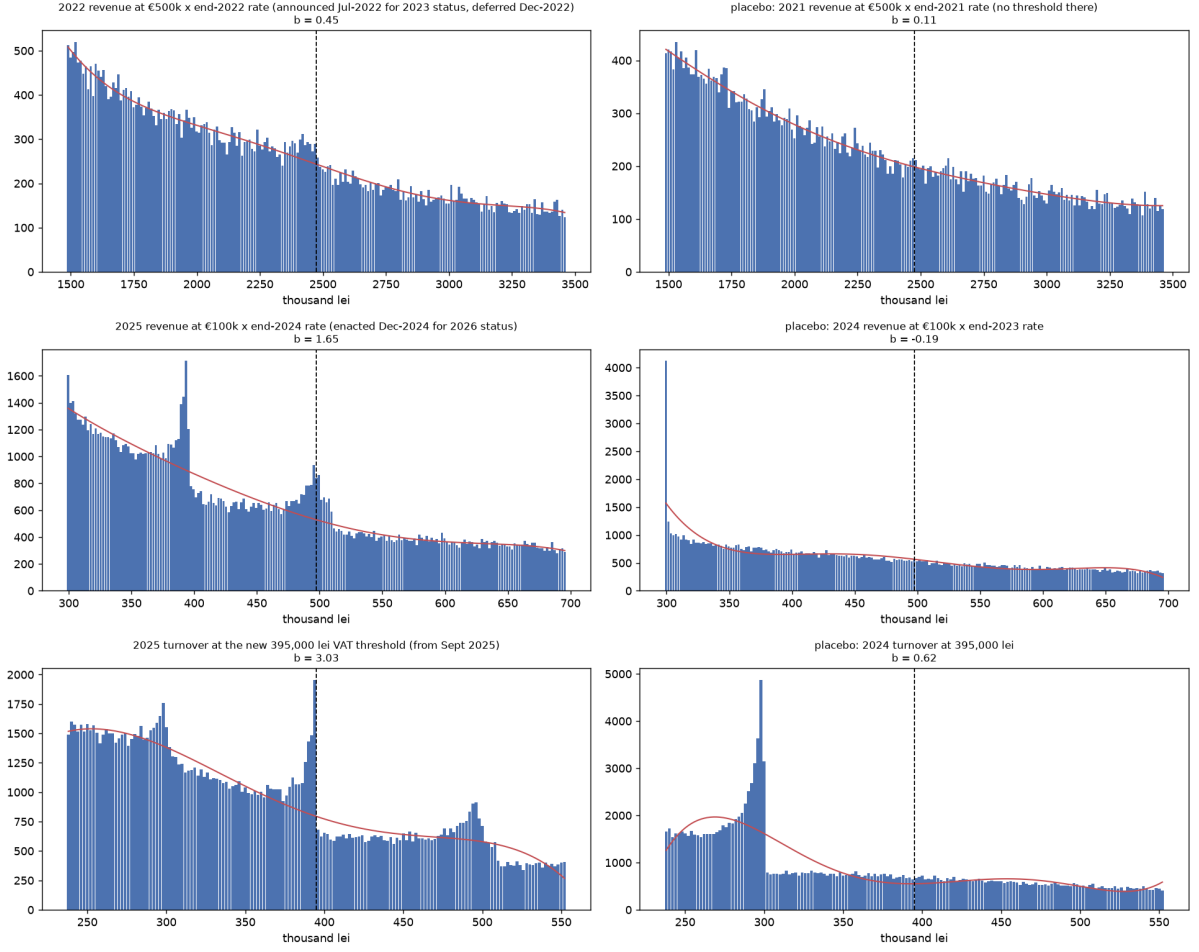


Figure 7: Revenue around announced or enacted future ceilings and the new VAT threshold, with placebo years. Top: 2022 revenue at €500,000, announced in July 2022 for 2023 status and deferred in December 2022, with 2021 placebo. Middle: 2025 revenue at €100,000 at the end-2024 rate, enacted December 2024 for 2026 status, with 2024 placebo. Bottom: 2025 turnover at 395,000 lei with 2024 placebo. Polynomial counterfactuals for reference.

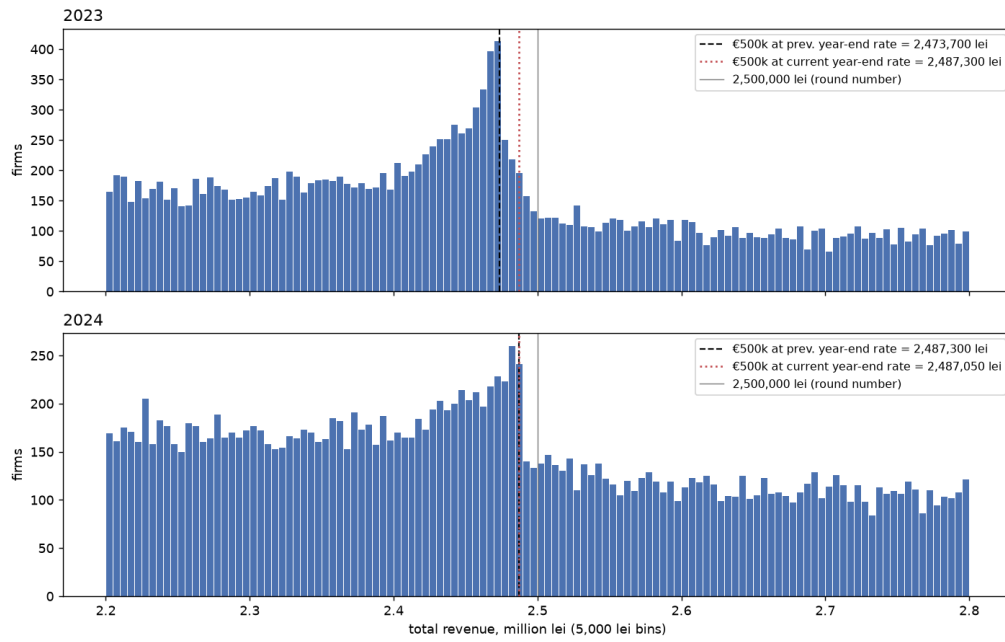


Figure 8: Total revenue in lei around €500,000, 2023 and 2024, 5,000-lei bins. The grey line marks 2,500,000 lei, the nearest round number.

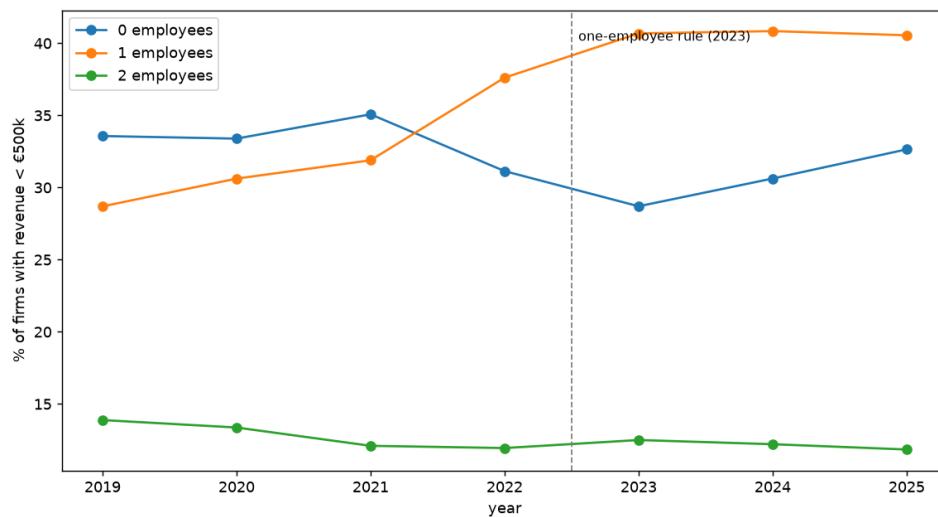


Figure 9: Share of firms with revenue below €500,000 reporting zero, one and two employees, 2019–2025.